Hesam Shafiei Nia

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SUMMARY

- Transportation Engineer with B.Sc. in Civil Engineering and two M.Sc. degrees in Transportation Engineering
- Proficient in traffic simulation (VISSIM, AIMSUN, Synchro, HCS), GIS (ArcGIS Pro), and data analysis (Python, SQL)
- Skilled at translating complex technical analyses into clear, actionable findings for engineering and planning applications
- Experienced in ML and data science: designing and building predictive models and data pipelines (Python, SQL, Git)
- Available to start: January 2027

TECHNICAL SKILLS

- **Traffic & Simulation:** VISSIM, Synchro, HCS, AIMSUN,
- **Design & GIS :** AutoCAD Civil 3D, ArcGIS Pro, ArcGIS Online, Microsoft Excel
- **Programming & Tools:** Python, SQL, R, MATLAB, Git/GitHub, Linux, PyTorch

EXPERIENCE

Graduate Research Assistant — Trajectory Reconstruction & Emissions Modeling *Dec. 2025 – Dec.2026*

University of Washington, Seattle — Advisors: Dr. Don MacKenzie & Dr. Cynthia Chen

- Designed a physics-informed deep autoencoder (PyTorch) to reconstruct ADS-B flight trajectory data, benchmarked against baseline models
- Built and validate a 3D spatial data analysis pipeline (Python, ArcGIS Pro) to model transportation system performance and emissions across 3 major airport hubs
- Collaborated with faculty advisors to align technical outputs with project goals; documented methodology and results in structured technical reports

Graduate Research Assistant — The State of Zero-Emissions Delivery in the U.S. *Fall 2024*

Urban Freight Lab (UFL), University of Washington, Seattle

- Analyzed transportation infrastructure needs, including electrification pathways and charging network requirements for commercial freight fleets; contributed findings through team-based research to a published industry report

Graduate Project — Traffic Signal Simulation—Pretimed vs. Actuated Control *Spring 2026*

University of Washington

- Design and built VISSIM Micro Simulation models for a real high-volume signalized intersection
- Designed a full NEMA 8-phase, 2-ring, 2-barrier actuated controller with inductive loop detectors and MUTCD-compliant pedestrian walk and FDW intervals; validated model outputs against HCM analytical baseline (Synchro, HCS)
- Documented and communicated where simulation and analytical methods diverge under congested demand conditions; project also included freeway microsimulation and ramp metering models

Graduate Thesis Research — Corridor Simulation & Design Analysis (under SRBIAU) *2017–2020*

- Developed a corridor-level traffic simulation model (AIMSUN) for a dense urban arterial, analyzing how lane configuration, access management, and pedestrian crossing modifications affected throughput, delay, and emissions; produced technical report with quantitative findings by scenario

EDUCATION

M.Sc. in Transportation Engineering *Dec. 2026*

University of Washington, Seattle, WA — GPA: 3.63/4.0

M.Sc. in Transportation Engineering *Sep. 2020*

Islamic Azad University Science and Research Branch (SRBIAU), Tehran, Iran — GPA: 16.60/20

B.Sc. in Civil Engineering *July 2017*

Ilam University, Ilam, Iran